WORKED SOLUTION

STEP III 1987, Question 7

Pure • Series for Arctangent

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q7

The key idea

The series for $\tan^{-1} t$ comes from integrating $1/(1+x^2)$, and $1/(1+x^2)$ is the sum of a geometric series with ratio $-x^2$. Stopping the geometric series after finitely many terms leaves an *exact* remainder, and integrating that remainder gives the error term. On $[0,1]$ the factor $1/(1+x^2)$ lies between $\frac{1}{2}$ and 1, which traps the error between two easy integrals.

Solution

The identity

For any real x, the finite geometric series with first term 1 and ratio $-x^2$ gives

$$\sum_{r=0}^{n}(-x^2)^r = \frac{1-(-x^2)^{n+1}}{1+x^2},$$

which is valid because the ratio $-x^2$ is never equal to 1. Rearranging,

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots + (-1)^n x^{2n} + (-1)^{n+1}\frac{x^{2n+2}}{1+x^2}.$$

This is an exact identity, not an approximation. Integrate both sides from 0 to t. Since $\int_0^t \frac{\mathrm{d}x}{1+x^2} = \tan^{-1} t$ and $\int_0^t x^{2r}\,\mathrm{d}x = \frac{t^{2r+1}}{2r+1}$,

$$\tan^{-1} t = t - \frac{t^3}{3} + \frac{t^5}{5} - \cdots + \frac{(-1)^n t^{2n+1}}{2n+1} + (-1)^{n+1}\int_0^t \frac{x^{2n+2}}{1+x^2}\,\mathrm{d}x,$$

as required. This holds for every real t.

The bounds

Write $S_n(t) = \sum_{r=0}^{n} \frac{(-1)^r t^{2r+1}}{2r+1}$. By the identity,

$$\left|\tan^{-1} t - S_n(t)\right| = \int_0^t \frac{x^{2n+2}}{1+x^2}\,\mathrm{d}x,$$

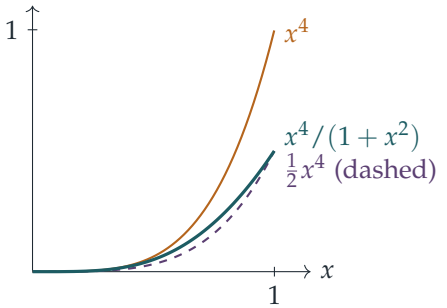
where we may drop the modulus on the right because, for $t \geq 0$, the integrand is non-negative and the integral runs forwards.

Ask yourself How can we estimate an integral we can't evaluate in closed form?

Bound the awkward part of the integrand by constants. For $0 \leq x \leq 1$ we have $1 \leq 1 + x^2 \leq 2$, so

$$\tfrac{1}{2}x^{2n+2} \leq \frac{x^{2n+2}}{1+x^2} \leq x^{2n+2}.$$

The picture shows this for $n = 1$: the integrand (teal) is squeezed between x^4 and $\frac{1}{2}x^4$.



Integrating these inequalities from 0 to t, which preserves them because $t \geq 0$, gives

$$\frac{t^{2n+3}}{2(2n+3)} \leq \left|\tan^{-1} t - S_n(t)\right| \leq \frac{t^{2n+3}}{2n+3} \qquad (0 \leq t \leq 1).$$

That is the required result.

The series for π

Put $t = 1$, where $\tan^{-1} 1 = \frac{\pi}{4}$ and $S_n(1) = \sum_{r=0}^{n} \frac{(-1)^r}{2r+1}$. The upper bound gives

$$\left|\pi - 4\sum_{r=0}^{n} \frac{(-1)^r}{2r+1}\right| \leq \frac{4}{2n+3} \to 0 \quad \text{as } n \to \infty,$$

so $4\sum_{r=0}^{n} \frac{(-1)^r}{2r+1} \to \pi$.

Ask yourself Which of the two bounds tells us the approximation is *bad*?

The lower bound does: it says the error can't be small. Multiplying the left-hand inequality at $t = 1$ by 4,

$$\left| \pi - 4 \sum_{r=0}^{n} \frac{(-1)^r}{2r+1} \right| \geq \frac{4}{2(2n+3)} = \frac{2}{2n+3}.$$

If $n \leq 98$ then $2n + 3 \leq 199 < 200$, so $\frac{2}{2n+3} > \frac{2}{200} = 10^{-2}$. Hence the error is at least 10^{-2} (in fact strictly more) for every $n \leq 98$. Since the error is below $4/(2n+3)$, a hundred terms buy only about two decimal places, so this series is a very slow way to compute π.

Checks

- $n = 0$, $t = 1$: the bounds say $\frac{1}{6} \leq \left| \frac{\pi}{4} - 1 \right| \leq \frac{1}{3}$. Indeed $1 - \frac{\pi}{4} \approx 0.215$.
- Numerically, with $n = 98$ the error is $|\pi - 4S_{98}(1)| \approx 0.01010$, just above 10^{-2}. With $n = 99$ it is about 0.0099998, just below. So 98 is exactly the cut-off. We also checked the identity by quadrature for random t and n, and the two-sided bound for thousands of random $t \in [0, 1]$.

Common slips

- Expanding $1/(1 + x^2)$ as an *infinite* series and integrating term by term. That needs $|x| < 1$ and a justification for swapping sum and integral. The finite version with an exact remainder avoids both.
- Using the upper bound to argue that the error is large. Only the lower bound can do that.

Beyond STEP

This is Taylor's theorem with an integral form of the remainder, worked out by hand for one function. The series at $t = 1$ is the Leibniz (or Madhava–Gregory) series. Its error of roughly $1/(2n)$ is typical of alternating series whose terms decrease slowly. Faster formulae, such as Machin's $\frac{\pi}{4} = 4\tan^{-1}\frac{1}{5} - \tan^{-1}\frac{1}{239}$, apply the same series at small t, where t^{2n+3} shrinks rapidly. For remainders and convergence done rigorously, see Abbott, *Understanding Analysis*.

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